

A direct proof of Morley's theorem

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Most direct proofs of Morley's theorem are very involved. Dick Hogg's neat geometrical proof (Note 67.3) circumvents the problem with its slightly indirect approach, but is very pleasing, none the less. It may however not be generally known that there is a direct straightforward proof using complex numbers which is simple to those familiar with the technique involved; indeed the proof is almost inevitable, once the right start has been made.

1. Turns

Everyone is familiar with the subset $\mathbb{R}$ of the complex numbers $\mathbb{C}$ represented in the complex number plane (often called the Argand diagram because it was invented by Wessel!) by the x -axis. Not so familiar is the subgroup (under multiplication) whose elements are the numbers with unit modulus, represented by points on the circle U with centre 0 and radius 1. If t is such a number, and $\arg t = \theta$, then $t = \cos \theta + i \sin \theta$, and it is a familiar elementary result that the multiplicative group of the t 's is isomorphic to the additive group (mod 2π) of the numbers θ and also to the group of rotations about 0 through the angles whose anticlockwise radian measures are these numbers θ . For this reason we may call such a number a *turn*; another term which is nowadays not well known is the word "affix" of a point P in a plane for the complex number which P represents. If t is a turn, and t^* its conjugate, then $tt^* = |t|^2 = 1$, so that $t^* = 1/t$, and conversely. If p, q are the affixes of points P, Q of U , then q/p is the turn corresponding to the rotation through angle POQ , and p, p^* correspond to equal and opposite rotations. Turns are really more elegant and manageable than expressions involving angles.

2. Preliminaries

(a) Two important turns are $\omega = \cos(2\pi/3) + i \sin(2\pi/3)$ and its square (and conjugate) $\omega^2 = \cos(4\pi/3) + i \sin(4\pi/3)$ corresponding to rotations through $2\pi/3$ and $4\pi/3$. Evidently $\omega^3 = 1$ and $1, \omega, \omega^2$ are the roots of the equation $x^3 = 1$, so that

$$1 + \omega + \omega^2 = \text{sum of roots} = 0.$$

The points whose affixes are p, q, r form an equilateral triangle if and only if $(p - q)/(r - q)$ is a turn corresponding to a rotation of $\pm\pi/3$, i.e., if $p - q = -\omega^2(r - q)$ or $-\omega(r - q)$. This gives

$$p - q(1 + \omega^2) + \omega^2 r = 0 \quad \text{or} \quad p - q(1 + \omega) + \omega r = 0$$

so that either

$$p + \omega q + \omega^2 r = 0 \quad \text{or} \quad p + \omega^2 q + \omega r = 0$$

depending on whether the sense of description of the triangle pqr is anticlockwise or clockwise.

(b) The standard equation of a straight line can be taken as $p^*z + pz^* = 1$. If this is the chord joining the turns t, u , then the equation $p^*s + p/s = 1$ must have roots t, u . Consequently $p^*s^2 - s + p = 0$ is the same equation as $s^2 - s(t + u) + tu = 0$, giving

$$\frac{p^*}{1} = \frac{1}{t + u} = \frac{p}{tu},$$

and the chord is $z + tuz^* = t + u$.

3. Morley's theorem

We take the circumcircle of the triangle ABC to be U , and the affixes of A, B, C to be a^3, b^3, c^3 , where a, b, c are turns in the minor arc from 1 to ω . Then one-third of the positive rotation from A to B is given by b/a ; one-third of that from B to C is given by c/b . However a/c corresponds to one-third of the rotation through the *negative* angle COA which contains B (see Fig. 1). We need one-third of the positive rotation from OC to OA , which is $2\pi + \angle COA$ so defined; so this is given by $\omega a/c$. The points of trisection of the consecutive arcs AB, BC, CA in Fig. 1 are therefore given by the turns $a^2b, ab^2; b^2c, bc^2; \omega ac^2, \omega^2 a^2c$.

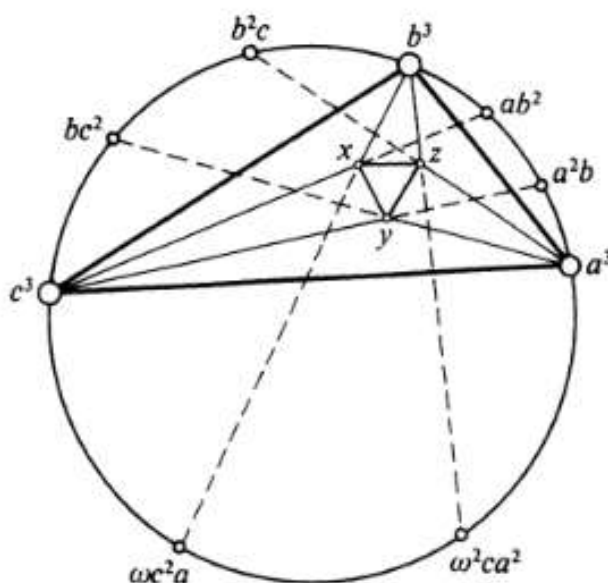


FIGURE 1.

The point x is the meet of the chords

$$x + ab^2c^3x^* = ab^2 + c^3$$

$$x + \omega ab^3c^2x^* = \omega ac^2 + b^3.$$

Subtracting and dividing by $(c - \omega b)$ gives at once

$$\begin{aligned} a^2b^2c^2x^* &= -\omega a^2(c + \omega b) + ac^2 + \omega abc + \omega^2 ab^2 \\ &= \omega^2(ab^2 - a^2b) + ac^2 - \omega a^2c + \omega abc. \end{aligned}$$

Similarly

$$a^2b^2c^2y^* = a^2b - ab^2 + bc^2 - b^2c + abc \quad (*)$$

and

$$a^2b^2c^2z^* = a^2c - \omega^2 ac^2 + \omega(b^2c - bc^2) + \omega^2 abc.$$

Therefore $x^* + \omega^2 y^* + \omega z^* = 0$; i.e., $x + \omega y + \omega^2 z = 0$, and the triangle xyz is equilateral.

Morley was familiar with this method of dealing with the properties of a triangle—indeed, he probably originated it. But he did not use it to prove his theorem, which he derived from consideration of the centres of cardioids touching the sides of the triangle [1]. He did however point out that an angle has three pairs of trisectors, and that the centres of cardioids touching two sides of the triangle and bitangent to the third side will lie on a set of 27 lines, 9 in each of three directions making angles of $2\pi/3$ with each other. It was left to W. J. Dobbs [2] to examine the system in detail. Our method simplifies this task, as I shall now show.

4. Extension to the full set of trisectors

Just as an angle has two bisectors at right angles to one another, so it has three pairs of trisectors at angles of $2\pi/3$, $4\pi/3$ with one another. To see what happens, consider Fig. 2. Suppose the rotation from AB to AC is given by the angle 3θ ; the turn corresponding to 2θ we have already seen to be c/b ($\angle BOC = 6\theta$). We can also rotate from AB to AC through an angle $2\pi + 3\theta$; the corresponding angle at the centre will be $6\theta + 4\pi$. One-third of this will be $2\theta + 4\pi/3$ and the corresponding turn will be $\omega^2 c/b$. The trisection points on U will therefore be given by the turns $\omega^2 cb^2$, $\omega c^2 b$. The chords from A to these points trisect the exterior angle $3\theta - \pi$ between AB and CA produced. Finally we can take the rotation from AB to AC through an angle $4\pi + 3\theta$. The trisection points are now ωcb^2 , $\omega^2 c^2 b$, and the chords trisect the reflex angle $3\theta - 2\pi$. Taking the three choices of trisectors at each vertex we obtain $3.3.3 = 27$ "Morley triangles" and $9 + 9 + 9 = 27$ vertices.

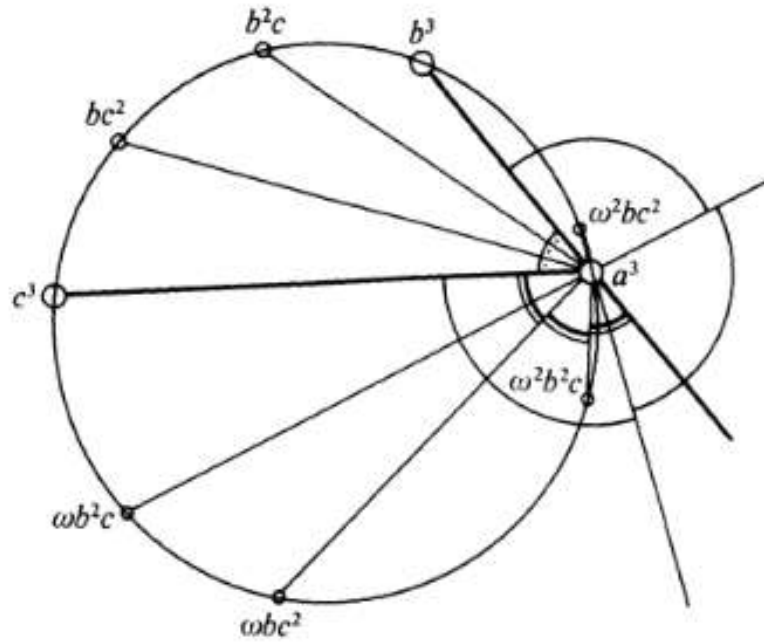


FIGURE 2.

5. The set of Morley points

We need a compact notation. The conjugate of equation (*) gives

$$y = b(c^2 + a^2) + abc - ca(c + a).$$

This expression is symmetrical in a and c ; let us call it $\left(b \middle| \begin{smallmatrix} c \\ a \end{smallmatrix} \right)$. Note that if $k = \omega$ or ω^2 , then $\left(ka \middle| \begin{smallmatrix} kb \\ kc \end{smallmatrix} \right) = \left(a \middle| \begin{smallmatrix} b \\ c \end{smallmatrix} \right)$. We also find that $x = \left(a \middle| \begin{smallmatrix} b \\ \omega^2 c \end{smallmatrix} \right)$, $z = \left(c \middle| \begin{smallmatrix} \omega a \\ b \end{smallmatrix} \right)$.

By repeating the calculations for the exterior trisectors we find that adjacent trisectors meet in pairs at the three points

$$\xi = \left(a \middle| \begin{smallmatrix} \omega^2 b \\ c \end{smallmatrix} \right), \quad \eta = \left(b \middle| \begin{smallmatrix} \omega^2 c \\ \omega a \end{smallmatrix} \right), \quad \zeta = \left(c \middle| \begin{smallmatrix} a \\ \omega b \end{smallmatrix} \right),$$

and similarly that the reflex trisectors meet in pairs at

$$X = \left(a \middle| \begin{smallmatrix} \omega b \\ \omega c \end{smallmatrix} \right), \quad Y = \left(b \middle| \begin{smallmatrix} \omega c \\ \omega^2 a \end{smallmatrix} \right), \quad Z = \left(c \middle| \begin{smallmatrix} \omega^2 a \\ \omega^2 b \end{smallmatrix} \right).$$

It is immediate that the triangles $\xi\eta\zeta$, XYZ are equilateral. Alternatively we can check that the ratio $(Y - Z)/(y - z)$ is the real number

$$(b - \omega c)(c - \omega^2 a)(a - \omega b) / \{(b - c)(c - \omega a)(a - b)\},$$

and that $(\eta - \zeta)/(y - z)$ is also real, so that the sides of the three triangles are parallel. There is however no reason why we should use the same set of trisectors for each of the angles A, B, C . Suppose BX meets Cx at x' , Bx meets CX at x'' , Bx meets $C\xi$ at ξ' , $B\xi$ meets Cx at ξ'' , $B\xi$ and CX meet at X' , and BX and $C\xi$ meet at X'' . Name the y and z points similarly. We then find that the affixes of the 27 points so obtained are as given in the table in §7. The last column of this table has yet to be explained.

6. Collinear points

It is straightforward algebra to verify that

$$(c-a)\omega^2 \left(b \middle| \begin{smallmatrix} c \\ a \end{smallmatrix} \right) + (\omega c - a) \left(c \middle| \begin{smallmatrix} \omega a \\ b \end{smallmatrix} \right) + (c - \omega a) \left(a \middle| \begin{smallmatrix} b \\ \omega c \end{smallmatrix} \right) = 0$$

and, if $p = \frac{a}{b} + \frac{b}{c} + \frac{c}{a}$, then

$$\begin{aligned} (p^* - p) \left(b \middle| \begin{smallmatrix} \omega^2 c \\ \omega^2 a \end{smallmatrix} \right) + (\omega p^* + p)(1 - \omega^2) \left(c \middle| \begin{smallmatrix} \omega a \\ b \end{smallmatrix} \right) \\ + \omega(\omega p - p^*) \left(a \middle| \begin{smallmatrix} b \\ \omega c \end{smallmatrix} \right) = 0 \end{aligned} \quad (**)$$

These equations demonstrate the collinearity of the four points

$$y = \left(b \middle| \begin{smallmatrix} c \\ a \end{smallmatrix} \right), \quad Y'' = \left(b \middle| \begin{smallmatrix} \omega^2 c \\ \omega^2 a \end{smallmatrix} \right), \quad z = \left(c \middle| \begin{smallmatrix} \omega a \\ b \end{smallmatrix} \right), \quad x' = \left(a \middle| \begin{smallmatrix} b \\ \omega c \end{smallmatrix} \right).$$

Making the permutation $b \rightarrow \omega^2 c \rightarrow a \rightarrow b$ in (**) brings in the point $Z' = \left(c \middle| \begin{smallmatrix} \omega^2 a \\ \omega b \end{smallmatrix} \right)$ on the same line, and the further permutation $a \rightarrow \omega^2 b \rightarrow c \rightarrow a$ adds finally the point $x'' = \left(a \middle| \begin{smallmatrix} \omega b \\ \omega^2 c \end{smallmatrix} \right)$, so that the six points y, z, x', x'', Y'', Z' are all collinear.

7. Group structure

The configuration of the points can best be elucidated by considering the structure of the group of operations on the expressions $\left(b \middle| \begin{smallmatrix} c \\ a \end{smallmatrix} \right)$. Let P be the cyclic permutation (abc) , so that $P \left(b \middle| \begin{smallmatrix} c \\ a \end{smallmatrix} \right) = \left(c \middle| \begin{smallmatrix} a \\ b \end{smallmatrix} \right)$. Let S multiply the third

element by ω^2 , while T multiplies the second element by ω , so that $S\left(b\left|\begin{smallmatrix} c \\ a \end{smallmatrix}\right.\right) = \left(b\left|\begin{smallmatrix} c \\ \omega a \end{smallmatrix}\right.\right)$ and $T\left(b\left|\begin{smallmatrix} c \\ a \end{smallmatrix}\right.\right) = \left(b\left|\begin{smallmatrix} \omega c \\ a \end{smallmatrix}\right.\right)$. Then P, S, T generate a non-Abelian group

Table of Morley points

	$\begin{smallmatrix} x \\ I \end{smallmatrix}$	$\begin{smallmatrix} y \\ R \end{smallmatrix}$	$\begin{smallmatrix} z \\ R^2 \end{smallmatrix}$	
x	$\left(a\left \begin{smallmatrix} b \\ \omega^2 c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} c \\ a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega a \\ b \end{smallmatrix}\right.\right)$	I
x'	$\left(a\left \begin{smallmatrix} b \\ \omega c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} c \\ \omega^2 a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega a \\ \omega^2 b \end{smallmatrix}\right.\right)$	S
x''	$\left(a\left \begin{smallmatrix} \omega b \\ \omega^2 c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega c \\ a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega^2 a \\ b \end{smallmatrix}\right.\right)$	T
ξ	$\left(a\left \begin{smallmatrix} \omega^2 b \\ c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega^2 c \\ \omega a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} a \\ \omega b \end{smallmatrix}\right.\right)$	$S^2 T^2$
ξ'	$\left(a\left \begin{smallmatrix} \omega^2 b \\ \omega^2 c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega^2 c \\ a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} a \\ b \end{smallmatrix}\right.\right)$	T^2
ξ''	$\left(a\left \begin{smallmatrix} b \\ c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} c \\ \omega a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega a \\ \omega b \end{smallmatrix}\right.\right)$	S^2
X	$\left(a\left \begin{smallmatrix} \omega b \\ \omega c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega c \\ \omega^2 a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega^2 a \\ \omega^2 b \end{smallmatrix}\right.\right)$	ST
X'	$\left(a\left \begin{smallmatrix} \omega b \\ c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega c \\ \omega a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} \omega^2 a \\ \omega b \end{smallmatrix}\right.\right)$	$S^2 T$
X''	$\left(a\left \begin{smallmatrix} \omega^2 b \\ \omega c \end{smallmatrix}\right.\right)$	$\left(b\left \begin{smallmatrix} \omega^2 c \\ \omega^2 a \end{smallmatrix}\right.\right)$	$\left(c\left \begin{smallmatrix} a \\ \omega^2 b \end{smallmatrix}\right.\right)$	ST^2

$$ST = TS, RST = STR, TR = RS^2, SR = RS^2 T, R^3 = S^3 = T^3 = I, RS = T^2 R, RT = ST^2 R$$

of order 27 in which every element except I is of order 3. S, T commute, and (ST) commutes with P . Then $PS = T^2P$, $PS^2 = TP$, $PT = ST^2P$, $SP = PS^2T$. The centre is $\{I, ST, S^2T^2\}$. The geometry of the situation is best expressed in terms of the operators $R = TP$ and $Q = PS$. Then the operations I, R, R^2 applied to any of the points give a directly equilateral triangle; the operations I, Q, Q^2 give an oppositely equilateral triangle.

The table has been drawn up to conform to Dobbs's notation for the points. The basic Morley triangle xyz is given in the first row, by applying I, R, R^2 to the affix of x . Succeeding rows are obtained by applying the operations in the last column to the affixes of x, y, z in the first row; the name of the x element is given in the first column. Because of the non-commutativity these rows do not (in general) form equilateral triangles. For example $\zeta' = T^2Ry$, but $RT^2y = R\eta' = Z'$.

A set of six collinear points is obtained by applying to any of the affixes the set of operations $I, ST^2, R, S^2TR, SR^2, TR^2$. This set is not of course a subgroup, as Lagrange's theorem shows at once, but consists of the subgroup $\{I, TR^2, RT^2\} = H$ and the left-coset RH . There are two 6-lines through each point; when the set is applied to three of the points on a line we get the same line each time, but the other three give the alternative 6-lines, which make angles of $\pi/3$ with the original line. The interested reader may also like to verify and explore the ramifications of the following facts: if K is the subgroup $\{I, SR, R^2S^2\}$ and L is the subgroup $\{I, T^2R, R^2T\}$, then

- (a) $HR = RL, HR^2 = R^2K, KR^2 = R^2L, LR = RK$;
- (b) $H + RH$ gives a set of 6 collinear points, as does $K + R^2K$ and $RL + R^2L$, and the three lines form an equilateral triangle.

8. Conclusions

To sum up, we have the following results, which have either been proved or are now easily verified:

1. The operations I, R, R^2 applied to the points in the first column give 9 directly equilateral Morley triangles.
2. The operations I, TR, SR^2 applied similarly give 9 oppositely equilateral Morley triangles.
3. The other 9 Morley triangles are not in general equilateral.
4. The operations I, S^2T, ST^2 applied to the points in rows 1, 2 and 5 of the table produce 9 equilateral triangles whose vertices are all x points, all y points, or all z points, e.g. $x'x''\xi$ is equilateral.
5. The operations I, S, S^2 or I, T, T^2 applied to any point in the table give sets of three points collinear with a vertex.
6. The operations $I, T^2S, R, TS^2R, TR^2, SR^2$ applied to the rows I, ST, S^2T^2 in the table give a set of 9 lines, each containing 6 points, whose intersections two by two give every point in the set.

These conclusions lead to the final diagram, (Fig. 3). which is taken from Dobbs's original paper. It is most easily drawn by starting with the triangle XYZ , noting that $\angle BXC = \angle CYZ = \angle BZY = A/3$, etc., as the reader is invited to verify.

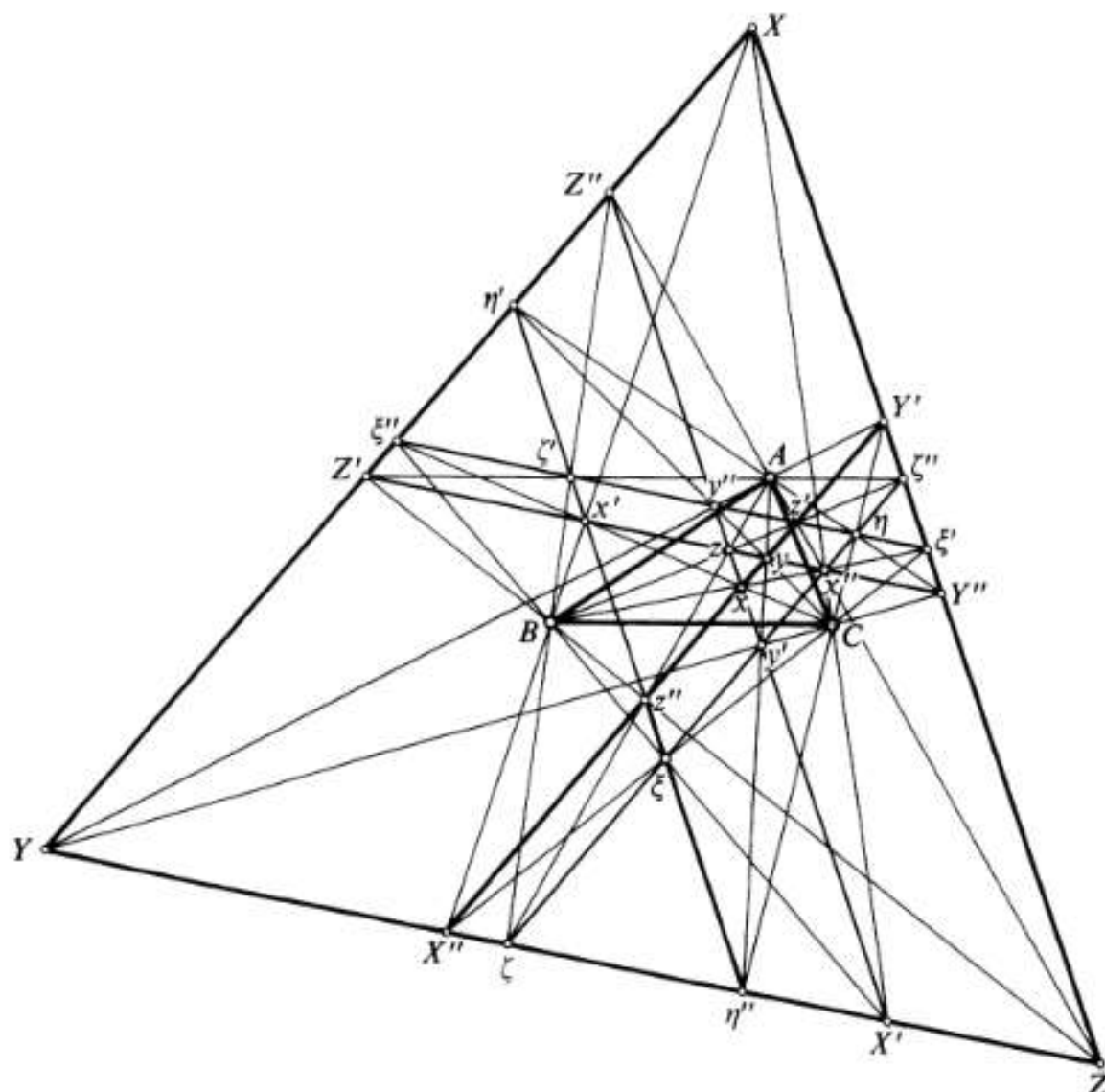


FIGURE 3.

References

1. E. and F. V. Morley, *Inversive geometry*.
2. W. J. Dobbs, Morley's triangle, *Mathl. Gaz.* XXII, 50

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Highly irregular

"Divide dough in half and roll each half into a sausage 8" (20 cm) long. Slice dough into $\frac{1}{4}$ " ($\frac{1}{2}$ cm) slices. Bake in a hot oven. (Makes about 48 Butterscotch Shorties)" From a recipe on an *All-Bran* packet (spotted by G. R. Addington).